

Diagnosis and Management of Hypocalcaemia

HYPOCALCAEMIA = adjusted Ca^{2+} <2.2 mmol/L

1. What is patient's calcium level?

Most patients are symptomatic when Ca^{2+} <1.9 but it depend on the rate of fall

2. How severe is hypocalcaemia?

NB depends on symptoms

Values are adjusted serum calcium

In emergency situations measure ionized calcium

Mild	1.9-2.1	☐
Moderate /severe	<1.9	☐

3. Are there any signs or symptoms of hypocalcaemia ?

Weakness, Muscle cramps, Paraesthesia

(usually fingers, toes and around mouth),

Tetany, Carpo-pedal spasm (wrist flexion and fingers drawn together).

Confusion, Seizures

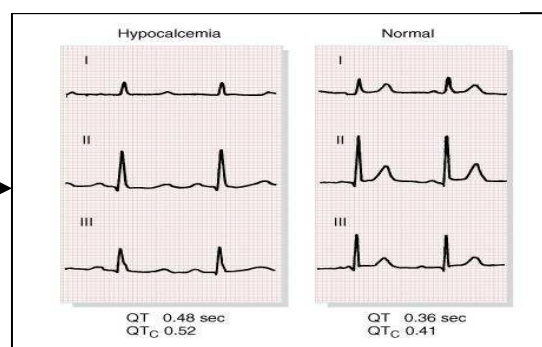
Behavioural change, Psychosis, Depression

Chvostek's sign, Trousseau's sign

Papilloedema

Heart failure

4. Are there any ECG changes of hypocalcaemia?



5. What are the main causes

The commonest cause of acute **symptomatic** hypocalcaemia in hospital practice is disruption of parathyroid gland function due to surgery (either **total thyroidectomy or parathyroidectomy**).

Hypocalcaemia in the context of blood transfusion is covered in 'Massive blood loss in adults' ([+ \(yha.com\)](http://yha.com))

Hypocalcaemia can also occur with vitamin D deficiency, in chronic kidney disease, acute pancreatitis and rhabdomyolysis, be secondary to hypomagnesaemia and drug related (cytotoxic drugs, bisphosphonates, denosumab)

6. Diagnosis of hypocalcaemia – the important questions

1. What is the serum **albumin**? Has the serum calcium been adjusted for serum albumin?

2. What is the serum **magnesium**? (If the Mg^{2+} is low, this suggests that the primary problem is **magnesium deficiency** – refer to **hypomagnesaemia guidance** [hypomagnesaemia-diagnosis-management \(yha.com\)](http://yha.com)).

3. What is **renal function**? (If renal failure is present, consider hypocalcaemia secondary to 1,25 dihydroxyvitamin D deficiency).

4 What are the **phosphate** and **alkaline phosphatase**?

A high phosphate level would be most suggestive of renal disease, but pseudohypoparathyroidism should also be considered

A low phosphate and high alkaline phosphatase is seen in malabsorption syndrome, severe vitamin D deficiency, osteomalacia and renal tubular acidosis

5. What is the **PTH**? (a low PTH in the presence of hypocalcaemia suggests hypoparathyroidism)

6. Therefore these laboratory tests are required to establish the cause of hypocalcaemia in patients that have not just undergone thyroid / parathyroid surgery

Serum calcium (total, adjusted for serum albumin; ionized in emergency cases), U&E, Phosphate, Mg^{2+} , Alkaline Phosphatase, PTH, Vitamin D (1,25 dihydroxyvitamin D is not routinely available)

Management of Acute Hypocalcaemia

Severe symptomatic hypocalcaemia is a **MEDICAL EMERGENCY**

Discuss hypocalcaemia following parathyroidectomy in the setting of renal failure with the Duty Renal Physician. Do not correct hypocalcaemia in the setting of rhabdomyolysis without seeking expert advice.

Administer IV calcium gluconate : Initially, give 10 ml 10% calcium gluconate in 50 ml of 5% glucose IV over 10min (with ECG monitoring) – see IV monograph	
Repeat 10% calcium gluconate 10 ml if symptoms not resolved. Measure serum calcium after each dose (ionized calcium can be obtained quickly). Patients with cardiac dysrhythmias or patients on digoxin therapy need continuous ECG	
A continuous calcium infusion may be required to maintain the plasma calcium in some patients. 100ml 10% calcium gluconate can be added to 1 L of 0.9% sodium chloride or 5% glucose solution and infused at 50 – 100 ml/hr Large volume calcium infusions should not be used in patients with end stage renal failure or who are on dialysis	
Titrate the rate of infusion to achieve normocalcaemia and continue until treatment of the underlying cause has been effective.	
Hazards of IV calcium administration: Uncommon but include local thrombophlebitis, cardiotoxicity, hypotension, metallic taste, flushing, nausea, vomiting and sweating.	
Treat the underlying cause; in post operative hypocalcaemia and other cases of hypoparathyroidism, this includes 1-alfacalcidol . Starting doses should be approximately 0.25 – 0.5 micrograms per day and check serum calcium 1 week post discharge and if satisfactory in 3 and 6 month's time. Use in renal patients should be discussed with a renal physician 1-alfacalcidol can be administered (at equivalent doses) intravenously if there are concerns about absorption or difficulties with oral drug administration Calcitriol is rarely used in the UK and not routinely available in the Trust.	
Vitamin D deficiency or hypomagnesaemia should be corrected (hypomagnesaemia-diagnosis-management (vha.com))	
“Mild” hypocalcaemia (Asymptomatic, >1.9 mmol/L) These guidelines do not apply to patients with renal failure	
Commence oral calcium supplements. Calcichew Forte (NB without vitamin D) 2 tablets BD (1 tablet = 2500mg) is the first line formulary preference. Sandocal™ 1000 , 2 tablets BD can be used in patients that have difficulty swallowing or other intolerance to calcichew. (Other alternatives should be discussed with pharmacy).	
If <u>post-thyroidectomy</u> and patient asymptomatic, repeat calcium 24 hours later When adjusted calcium is <u>> 2.1 mmol/L</u> , patient may be discharged and recheck calcium within one week If serum calcium remains between <u>1.9 and 2.1 mmol/L</u> increase Calcichew Forte to three BD. If patient remains in mild hypocalcaemic range beyond <u>72 hours</u> post operatively despite calcium supplementation, start 1-alfacalcidol 0.25 microgram/day with close monitoring.	
If vitamin D deficiency is the cause, treat with oral vitamin D .	
If hypomagnesaemia-related, identify and treat underlying cause of hypomagnesaemia	